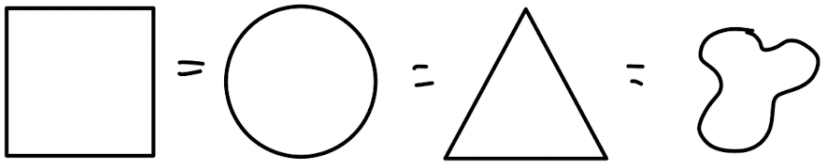
Introduction to Topology

The rough idea of topology is rubber sheet geometry. We aren't concerned with rigid notions such as distance. Instead we are concerned with properties like the number of holes, and how connected an object is.

Intuitive Idea about Topology : 2 objects are topologically equivalent if you can transform one to the other without breaking, making new holes, etc.



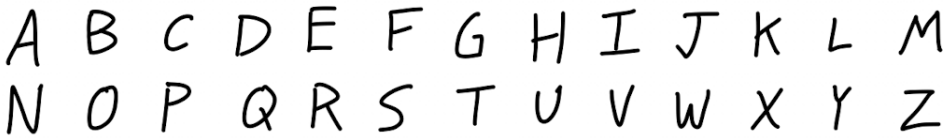
These are all examples of simple closed curves

Theorem : Jordan Curve Theorem

A simple closed curve divides the plane into an inside and an outside

Topological Classes

Consider the following alphabet (Note : the way each letter is drawn in important)



The topological classes are:

1. Simple closed curves: D, O
2. Simple open curves: C, G, L, M, N, S, U, V, W, Z
3. Simple closed curve with 2 simple open curves coming out of 2 different points: A, R
4. Q
5. B
6. E, F, J, T, Y
7. H, I, K
8. P,
9. X

- One way to check is to see what happens when you remove a point

Topology in $\mathbb{R}$

(We will be using the usual Euclidean metric on $\mathbb{R}$ which says the distance between two numbers x and y is $|x-y|$)

For a general metric space (a space X with some distance on it) we can generalize $|x-y|$ to $d(P,Q)$, the distance between points P and Q in X.

Definition : Delta Neighbourhood

Given $\delta > 0$, the δ-neighbourhood of a point $x \in \mathbb{R}$ is denoted by $V_\delta(x) = (x-\delta, x+\delta) = \{y \in \mathbb{R} : |x-y| < \delta\}$

Example

Let $X = \mathbb{R}^2$, and use the Euclidean metric: $d(P,Q) = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$.

$V_\delta(P) = \{Q \in \mathbb{R}^2 : d(P,Q) < \delta\}$

What is $d(P,Q) = \delta$?

$$\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2} = \delta \Rightarrow (x_2-x_1)^2 + (y_2-y_1)^2 - \delta^2$$

Let $P = (1,2)$, $\delta = 3 \Rightarrow (x_2-1)^2 + (y_2-2)^2 = 3^2$

So, $V_\delta(P)$ is the interior of the circle.

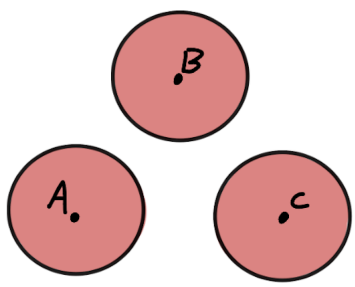
Consider an equilateral triangle (where each side has length 1), and let $S = \{A, B, C\}$, where A, B and C are the vertices of the triangle.

$$V_\delta(S) = \bigcup_{x \in S} V_\delta(x)$$

(The δ-neighbourhood of a set is the union of the δ-neighbourhoods of the points in the set

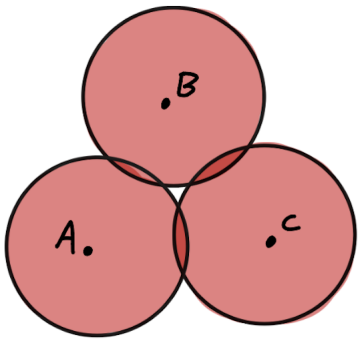
What do these δ-neighbourhoods look like? (It depends on the value of δ)

- $\delta < \frac{1}{2}$



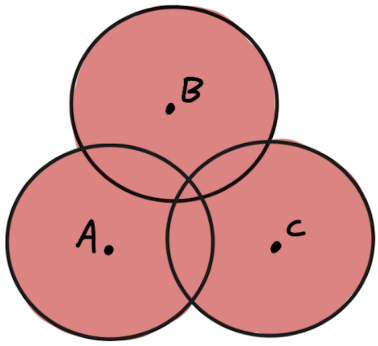
  This gives us 3 disjoint open sets

- $\delta > \frac{1}{2}$ (but not too big)



Here, the center point will not be covered

- $\delta > \frac{\frac{1}{2}}{\cos(30°)}$



Here, there will be no holes